Data Science Assignment Report
Trader Behavior vs. Market Sentiment

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1. Objective

The goal of this analysis was to explore the relationship between trader behavior and overall market sentiment. By analyzing profitability, trading volume, and risk-taking against the Fear & Greed Index, the objective was to uncover hidden patterns and identify data-driven strategic signals for smarter trading.

2. Methodology

1. **Data Loading:** Loaded historical_data.csv (containing 180,000+ individual trade records) and fear_greed_index.csv (a daily record of market sentiment).
2. **Data Cleaning:** The two datasets used different "time" formats. The trader data used a millisecond timestamp (a large number), while the sentiment data used a text-based date (e.g., "2024-10-27"). Both were converted into a standardized, clean datetime object.
3. **Data Merging:** To link the two datasets, a common trade_date key was created by stripping the specific time from the trader data. This allowed every single trade to be "tagged" with the market's sentiment for the day it occurred, creating a single, unified master dataset.
4. **Analysis:** The merged data was then grouped by the classification (Fear, Greed, Neutral, Extreme Greed) to analyze three core metrics:
   - **Profitability:** The sum (Total PnL) and average (Average PnL per Trade) of Closed PnL.
   - **Volume:** The total Number of Trades (count) and the Total USD Volume (sum of Size USD).
   - **Risk:** PnL Standard Deviation (as a measure of volatility) and the Average Loss on losing trades.

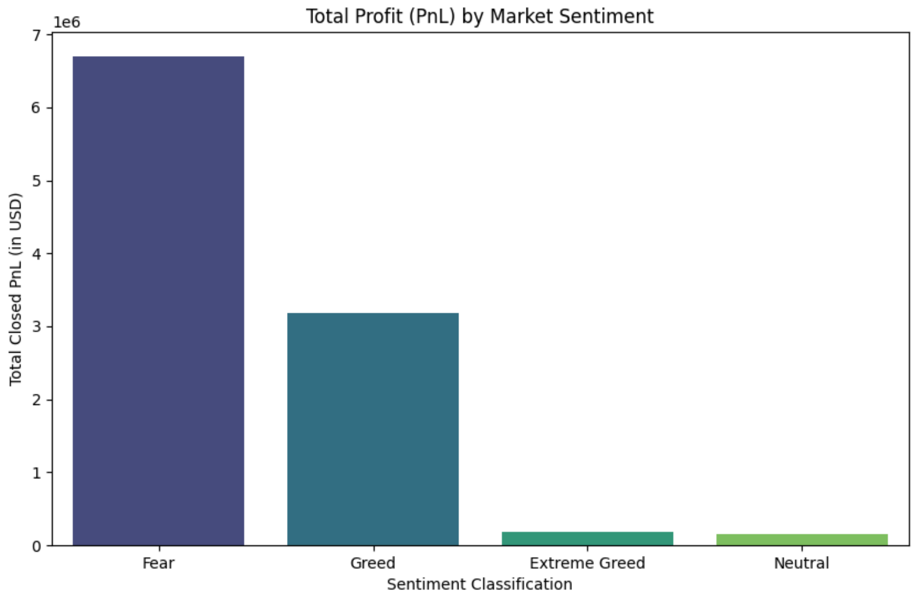
3. Key Findings

Analysis revealed four distinct "trading personalities" that emerge based on market sentiment.

Finding 1: "Fear" is a High-Volume Game

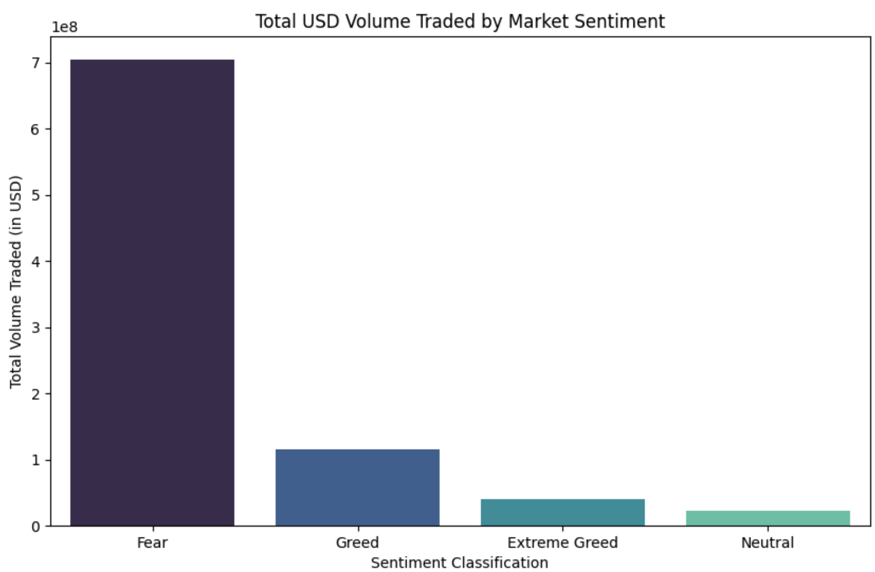
During 'Fear', the market is characterized by frantic activity. This high volume, not high individual profit, is the primary driver of market dynamics.

- **Highest Total Profit:** 'Fear' days generated the most cumulative profit at approximately **$6.7 million**.



- **Massive Volume:** This profit was not due to superior trades, but to sheer volume. 'Fear' days saw approximately **134,000 trades** totaling over **$704 million** in value—far exceeding all other categories combined.
- **Average Profit:** The average profit per trade was mediocre, at only **$50.05**.

This suggests a "grind" mentality, where profits are built from a large quantity of small- to medium-sized trades.

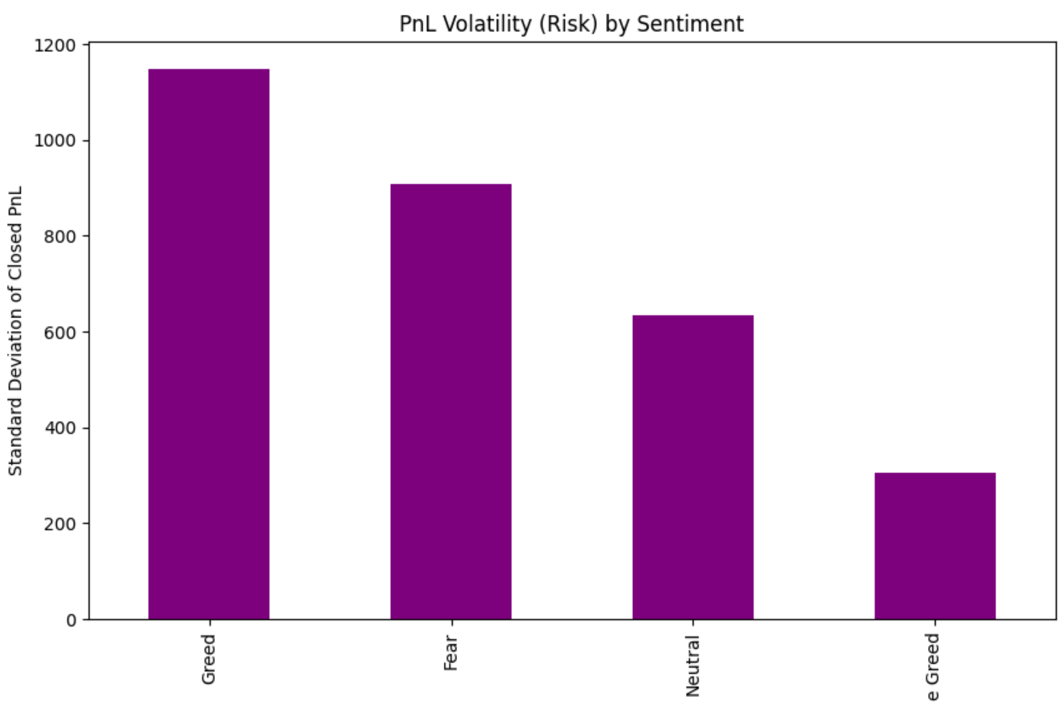


Finding 2: "Greed" is a High-Risk, High-Reward Game

During 'Greed', traders behave like "snipers," taking fewer, more selective, and significantly riskier bets.

- **Highest Average Profit:** 'Greed' days boasted the highest average profit per trade at approximately **$87.89**.
- **Highest Risk:** This period was by far the **most volatile** (PnL Standard Deviation of ≈1148). This indicates that trade outcomes had the wildest swings, leading to both the biggest wins and significant losses.
- **Low Volume:** This "sniper" approach was evident in the low volume, with only **36,000 trades** conducted.

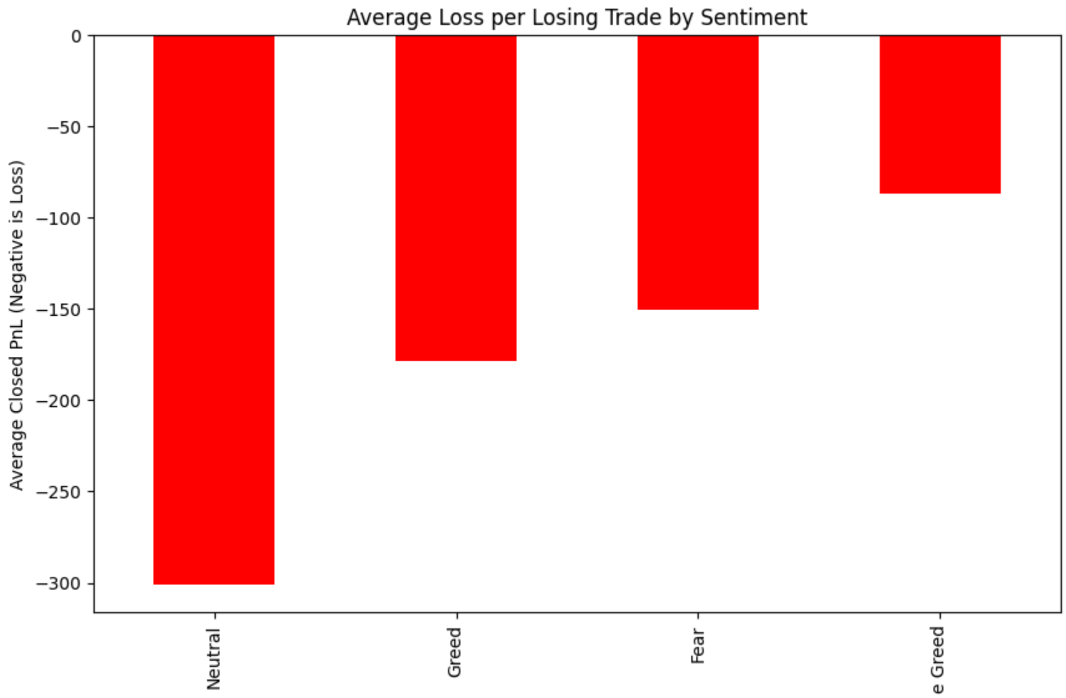
Traders are not grinding; they are making high-conviction, high-stakes bets.



Finding 3: "Neutral" is Deceptively Dangerous

'Neutral' days appear quiet but are the most punishing for traders who make a mistake.

- **Highest Average Loss:** When trades went wrong on 'Neutral' days, they were catastrophic. The average loss was -$301 per losing trade, which is 69% worse than 'Greed' (≈ -$178) and double that of 'Fear' (≈ -$150).
- **Low Reward:** This extreme downside risk was paired with the lowest average profit of any category, at only **$22.23**. This makes it, statistically, the worst risk/reward profile.



4. Conclusion & Strategic Signals

Trader behavior is not uniform; it shifts predictably with market sentiment.

- **"Fear" Strategy Signal:** This is a **volume game**. The signal suggests that profits are found in high-frequency activity. Traders are highly active, and while average wins are small, average losses are also contained, making it a "grind" environment.
- **"Greed" Strategy Signal:** This is a **precision game**. The signal suggests a "sniper" approach is rewarded. Traders are betting big (high risk/volatility) and winning big (high average PnL). This environment favors fewer, high-conviction trades.
- **"Neutral" Strategy Signal:** This is an **"avoid" signal**. The market is uncertain, and the data clearly shows that the cost of being wrong is higher than in any other period, while the potential rewards are minimal.